

Problem 9.3

Suppose that \$100 is deposited into a savings account, earning at a discount rate of 0.15% biweekly, at the beginning of year 2006.

- (a) Find the nominal annual discount rate.
- (b) Find the effective annual discount rate.
- (c) Find the amount of interest generated during the year.
- (d) Find the equivalent effective annual interest rate.
- (e) Find the equivalent nominal annual interest rate, convertible monthly.

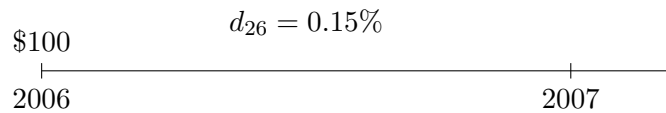
Solution

A biweekly period occurs every two weeks, so there are

$$\frac{52}{2} = 26$$

biweekly periods in one year. The biweekly discount rate is

$$d_{26} = 0.15\% = 0.0015.$$



(a)

The nominal annual discount rate convertible biweekly is

$$\begin{aligned} d^{(26)} &= 26(0.0015) \\ &= 0.039. \end{aligned}$$

Thus,

$$\boxed{d^{(26)} = 3.9\%}.$$

(b)

The effective annual discount rate d satisfies

$$1 - d = (1 - d_{26})^{26}.$$

Therefore,

$$\begin{aligned} d &= 1 - (1 - 0.0015)^{26} \\ &\approx 0.03827745. \end{aligned}$$

Thus,

$$\boxed{d \approx 3.827745\%}.$$

(c)

If A is the accumulated value at the end of the year, then

$$100 = A(1 - d),$$

so

$$\begin{aligned} A &= 100(1 - d)^{-1} \\ &\approx 100(1 - 0.03827745)^{-1} \\ &\approx 103.9800928. \end{aligned}$$

The amount of interest generated is

$$\begin{aligned} I &= 103.9800928 - 100 \\ &\approx 3.9800928. \end{aligned}$$

Thus,

$$\boxed{I \approx \$3.98}.$$

(d)

The equivalent effective annual interest rate is related to the effective annual discount rate by

$$i = \frac{d}{1 - d}.$$

Hence,

$$\begin{aligned} i &= \frac{0.03827745}{1 - 0.03827745} \\ &\approx 0.03980093. \end{aligned}$$

Thus,

$$\boxed{i \approx 3.980093\%}.$$

(e)

Let $i^{(12)}$ be the equivalent nominal annual interest rate convertible monthly. Then

$$1 + i = \left(1 + \frac{i^{(12)}}{12}\right)^{12}.$$

Solving for $i^{(12)}$,

$$\begin{aligned} i^{(12)} &= 12 \left((1 + i)^{1/12} - 1 \right) \\ &= 12 \left((1 + 0.03980093)^{1/12} - 1 \right) \\ &\approx 0.03909282. \end{aligned}$$

Therefore,

$$\boxed{i^{(12)} \approx 3.909282\%}.$$